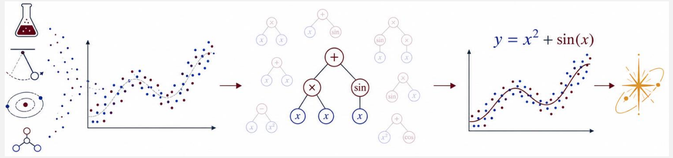
VaSST: Variational Inference for Symbolic Regression using Soft Symbolic Trees

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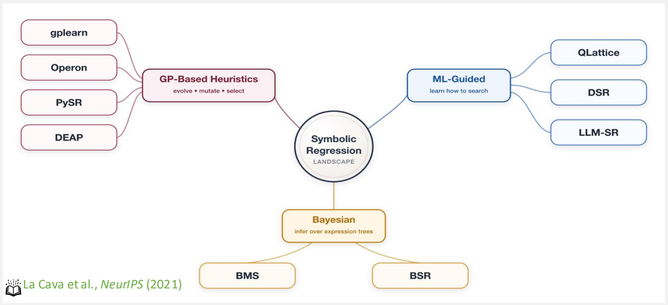
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Symbolic Regression for Scientific Discovery

- **Why:** Scientific discovery needs explicit equations that reveal mechanisms-not just accurate predictions.
- **What:** Data-driven learning of concise, interpretable mathematical expressions.



Symbolic Regression Landscape



La Cava et al., *NeurIPS* (2021)

Pitfalls

- **GP-based:** Expensive, initialization-sensitive, unwieldy formulas.
- **ML-guided:** Data-hungry, noise-sensitive, prompt-specific.
- **Bayesian:** scaling difficulty, ad hoc prior, poor MCMC mixing.

Why VaSST? Our Contributions

VaSST turns combinatorial equation search into scalable probabilistic learning—discovering interpretable scientific laws with quantified symbolic uncertainty.

- **Variational Inference** — Uncertainty over structures and model parameters
- **Soft Symbolic Trees** — Distributions over operators and features
- **Scalable Inference** — Gradient-based variational optimization
- **Posterior-Aware Discovery** — Ranks multiple plausible, concise equations
- **Empirically Strong** — Recovery • stability prediction • speed

VaSST Github Repo: *github.com/Roy-SR-007/VaSST*

The VaSST Model

Symbolic Ensemble Component

$$y = \beta_0 + \beta_1 \; \underset{x_1 \quad x_2}{\times} \; + \; \beta_2 \; \underset{x_3}{\sin} \; + \; \ldots \; + \; \varepsilon$$

ε is Gaussian noise with variance σ^2 ; $(\beta, \sigma^2) \sim$ conjugate Normal-Inverse-Gamma prior

Prior over full binary tree skeleton

Depth-dependent split prior

$$e_{j\zeta} \sim \mathrm{Ber}(p_\zeta = \alpha(1+d_\zeta)^{-\delta})$$

Chipman et al., *JASA* (1998)

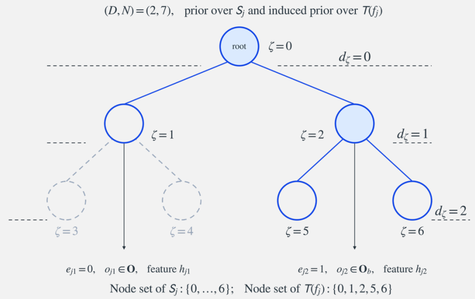
Operator and feature assignments

$$o_{j\zeta} \mid \mathbf{w}_{\mathrm{op}} \sim \mathrm{Cat}(\mathbf{w}_{\mathrm{op}})$$

$$h_{j\zeta} \mid \mathbf{w}_{\mathrm{ft}} \sim \mathrm{Cat}(\mathbf{w}_{\mathrm{ft}})$$

Weight priors

$$\mathbf{w}_{\mathrm{op}} \sim \mathrm{Dir}(\eta_{\mathrm{op}}) \, , \mathbf{w}_{\mathrm{ft}} \sim \mathrm{Dir}(\eta_{\mathrm{ft}}) \, .$$



$$\Pi(\mathsf{S}_j \mid \mathbf{w}_{\mathrm{op}}, \mathbf{w}_{\mathrm{ft}}) = \prod_{\zeta \in \mathsf{Z}_D} \Pi(e_{j\zeta}) \Pi(o_{j\zeta} \mid \mathbf{w}_{\mathrm{op}}) \Pi(h_{j\zeta} \mid \mathbf{w}_{\mathrm{ft}})$$

$$\Longrightarrow \Pi(\mathsf{T}(f_j) \mid \mathbf{w}_{\mathrm{op}}, \mathbf{w}_{\mathrm{ft}}) = \prod_{\zeta \in \mathsf{Z}(f_j)} \Pi(e_{j\zeta}) \Pi(o_{j\zeta} \mid \mathbf{w}_{\mathrm{op}}) \Pi(h_{j\zeta} \mid \mathbf{w}_{\mathrm{ft}})$$

The VaSST Pipeline

Mean-field Variational Inference $q_\phi(\Theta) = q_\phi(\mathbf{w}_{\mathrm{op}}) q_\phi(\mathbf{w}_{\mathrm{ft}}) \prod_{j=1}^{K} \prod_{\zeta \in Z_D} q_\phi(e_{j\zeta}) q_\phi(o_{j\zeta}) q_\phi(h_{j\zeta})$

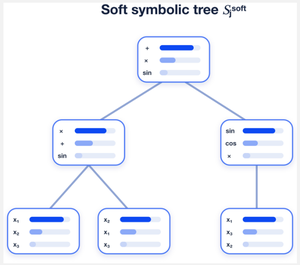
$\Theta \rightarrow$ symbolic tree parameters, $\phi \rightarrow$ variational parameters

Soft Symbolic Trees

Binary Concrete $e_{j\zeta} \rightarrow \tilde{e}_{j\zeta}$

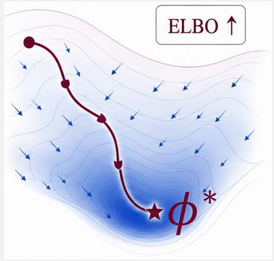
Gumbel Softmax $o_{j\zeta} \rightarrow \tilde{o}_{j\zeta}$; $h_{j\zeta} \rightarrow \tilde{h}_{j\zeta}$

Maddison et al., *ICLR* (2017); Jang et al., *ICLR* (2017)



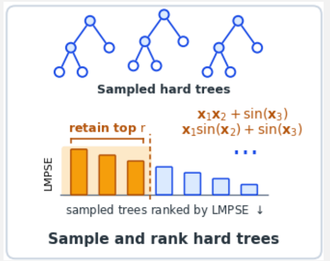
Evidence Lower Bound

$$\widehat{\mathcal{E}}(\phi) = \frac{1}{S} \sum_{s=1}^{S} \log p\left(\mathbf{y} \mid \mathbf{T}_{\mathrm{soft}}^{(s)}\right) - \mathrm{KL}[q_\phi(\Theta) \,\|\, \Pi(\Theta)] \, .$$

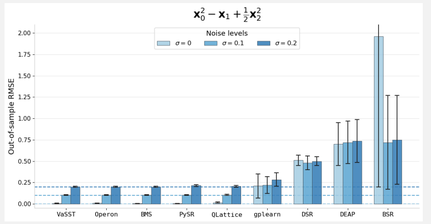


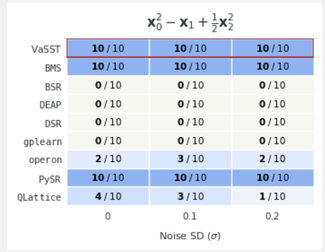
Posterior-aware Ranking

$$\mathrm{LMPSE}(\mathbf{T}) = \log \Pi(\mathbf{T} \mid \mathbb{D}_n) = \log p(\mathbf{y} \mid \mathbf{T}) + \log \Pi(\mathbf{T}).$$



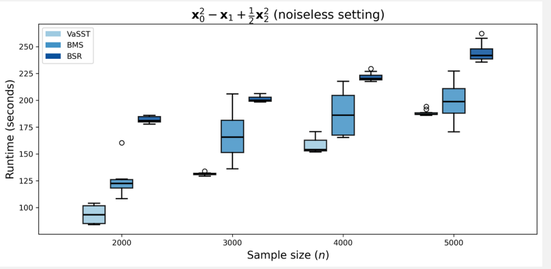
VaSST In Action





VaSST balances predictive accuracy with structural recovery

Computational scalability of VaSST against Bayesian competitors



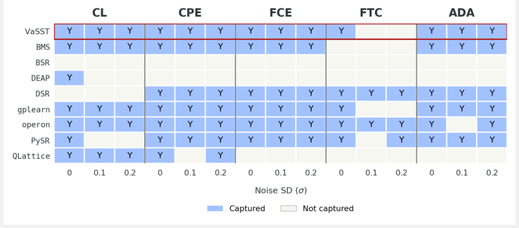
Learning Feynman Equations

Udrescu & Tegmark, *Science Advances* (2020)

$$F = 0.08 \frac{q_1 q_2}{\varepsilon r^2} \quad \text{(CL)}; \quad \Delta U = G m_1 m_2 \left(\frac{1}{r_2} - \frac{1}{r_1}\right) \quad \text{(CPE)};$$

$$F = q\left(E_f + vB\sin\theta\right) \quad \text{(FCE)}; \quad P = \frac{\kappa A (T_2 - T_1)}{d} \quad \text{(FTC)};$$

$$f = \beta^\dagger \left(1 + \alpha^\dagger \cos\theta\right) \quad \text{(ADA)}.$$



Feynman equation recovery of VaSST and competitors across different noise levels

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arXiv preprint: *https://arxiv.org/abs/2602.23561*